

ETHER TERMINAL LLC

Sovereign Cryptographic Intelligence & Autonomous Operating Systems

NORTH CAROLINA SOS FILE NO. 3300608
HEADQUARTERS: GREENVILLE, NC 27834
TEL: (888) 861-9020 | WEB: ETHER-TERMINAL.COM
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SOVEREIGN ETHER OS // ENTERPRISE PILOT DEPLOYMENT & ONBOARDING CHECKLIST

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Target Audience: Carrier Project Sponsors, Lead Solutions Architects, Fleet Operations Directors, Chief Risk Officers

System Target: Sovereign Ether OS (v1.0.0-institutional-final)

Deployment Duration: 30 Calendar Days (Day-0 Bench Readiness to Day-30 Production Transition)

1. Executive Summary & Deployment Objectives

This runbook defines the structured, stage-gated implementation roadmap for commissioning Sovereign Ether OS across an enterprise pilot fleet. The pilot is engineered to validate five non-negotiable institutional benchmarks without risking vehicle warranty invalidation or enterprise network compromise:

- 1. Zero CAN-Bus Write-Back: 100% passive diagnostic bus sniffing (listen-only on kernel mode) under SAE J1939 and ISO 14229 UDS.
- 2. Deterministic Dual-Consensus WASM: Flawless isolation and byte-identical consensus between Cranelift and V8 runtimes.
- 3. Edge Zero-PII Compliance: Complete in-memory redaction of driver records under DPPA (18 U.S.C. § 2721) prior to bundle export.
- 4. Daubert-Admissible Optical Forensics: Sub-120ms verification of physical CMOS Bayer CFA demosaicing peaks at (π, π) to eliminate synthetic generative media fraud.
- 5. Real-Time Institutional Solvency Automation: Deterministic 4-tier reinsurance waterfalls (NAIC Schedule F) and ISO 20022 parametric settlement dispatch.

30-DAY PILOT TIMELINE				
Phase 0 (Days -7..0)	Phase 1 (Days 1..5)	Phase 2 (Days 6..12)	Phase 3 (Days 13..20)	Phase 4/5 (21..30)
Digital Software	Cryptographic IAM	Fleet Installation	Carrier WASM Logic	Solvency & Go-Live
& Air-Gap Staging	& WireGuard Peering	& Passive Sniffing	& Optical Forensics	Sign-Off & Audit

2. Phase 0: Pre-Deployment & Air-Gap Environment Readiness (Days -7 to 0)

Before executing the pilot at carrier facilities, verify the digital delivery packages, cryptographic baseline, and host workstation readiness.

- [] 0.0 Digital Delivery & Wire Settlement Verification
- Verify receipt of digital air-gapped delivery package (macOS DMG installer / ZIP distribution with SHA-256 cryptographic attestation).

- **\$0.00 Hardware CapEx:** Zero new hardware required—executes 100% locally on carrier's existing enterprise workstations and computing infrastructure.
- Confirm presence of pre-loaded air-gapped forensic models and local cryptographic verification keys.
- Confirm Accounts Payable wire settlement received with matched Wire Memo Code (`PILOT-ET-*****`).
- [] **0.1 Host Workstation & Interface Verification**
- Verify host workstation meets enterprise minimums (macOS 13.0+ Ventura/Sonoma/Sequoia or Windows 10/11 64-bit).
- Verify tamper-resistant local execution and zero external network egress dependencies.
- For passive CAN vehicle auditing: Confirm presence of standard passive OBD-II / SAE J1939 breakout cable with hardware-clipped CAN Tx lines (`listen-only`).
- [] **0.2 Non-Volatile Memory (NVRAM) & Hardware Clock Audit**
- Verify battery-backed monotonic RTC hardware clock with hardware lease-clock enforcement.
- Execute NVRAM integrity check: Confirm zero-tolerance anti-rollback counter is initialized to zero: `bash nvrwam -p | grep "anti_rollback_monotonic_counter"`
- Verify 10-day offline grace period parameter in `EnterpriseCustomConfig.swift` (`offlineLicenseGracePeriodDays = 10`).
- [] **0.3 Network Loopback & Attack Surface Lockdown**
- Boot the edge unit in isolated bench environment (zero external WAN/LAN cables).
- Execute local port audit verifying zero public network daemons: `bash sudo netstat -tulnp | grep -v "127.0.0.1"`
- Confirm that REST (`127.0.0.1:8000`) and WebSocket (`127.0.0.1:8765`) are strictly bound to loopback.
- Verify local SSD partition is encrypted under `AES-256-XTS-FIPS-140-3` .

3. Phase 1: Cryptographic Identity & WireGuard Peering (Days 1 to 5)

Establish the zero-trust carrier mesh network and seed tenant cryptographic keys.

- [] **1.1 Hardware Enclave Ed25519 Key Generation**
- Trigger onboard Secure Element (Apple Secure Enclave or ATECC608B HSM) to generate the device identity keypair:
 - Public Key: Hardware Identity Public Key (attested to Carrier Root CA).
 - Private Key: Non-exportable, hardware-bound silicon key.
- Record the device SHA-256 fingerprint in the carrier pilot inventory database.
- [] **1.2 Carrier WireGuard Point-to-Point Tunnel Configuration**
- Establish WireGuard interface (`wg0`) listening exclusively on high UDP port: ```ini [Interface] Address = 10.68.0.24/32 PrivateKey = ListenPort = 51820`
`[Peer] PublicKey = Endpoint = vpn.carrier-domain.internal:51820 AllowedIPs = 10.68.0.1/32`
`PersistentKeepalive = 25 `` - Perform ChaCha20-Poly1305 packet authentication test over isolated LTE/5G APN.`
- [] **1.3 Tenant Custom Configuration Provisioning**
- Deploy `EnterpriseCustomConfig.swift` with verified carrier parameters:

- `tenantID` : "TENANT-CORP-US-001" (e.g., Sovereign Risk Syndicate AIG).
- `autoSettlementLimit` : \$50,000.00 (Straight-Through Processing ceiling).
- `highSeverityExecGateThreshold` : \$750,000.00 (Mandatory Executive Lock).
- `wasmDualConsensusEnforced` : `true` .

4. Phase 2: Fleet Vehicle Installation & Passive CAN Sniffing (Days 6 to 12)

Install units across the designated 20-vehicle pilot cohort (passenger sedans, delivery vans, Class 8 heavy trucks).

- [] **2.1 Physical OBD-II / J1939 Harness Integration**
- Connect harness to standard vehicle diagnostic port:
 - **Pins 6 & 14:** High-speed CAN differential pair (CAN-H / CAN-L).
 - **Pins 4 & 5:** Chassis and signal ground.
 - **Pin 16:** Unswitched battery positive (+12V / +24V DC).
- Verify that Transmit Pin (Tx) is physically absent or isolated via optocoupler with reverse-bias protection.
- [] **2.2 SocketCAN Passive Listening Verification**
- Initialize SocketCAN interface in kernel-enforced listen-only mode: `bash sudo ip link set can0 type can bitrate 500000 listen-only on sudo ip link set can0 up`
- Verify `listen-only on` flag using `ip -details link show can0` .
- For heavy commercial trucks, switch to SAE J1939 250 kbps profile: `bash sudo ip link set can0 type can bitrate 250000 listen-only on`
- [] **2.3 Zero ECU Interference & OEM Factory Warranty Audit**
- Run an external OEM diagnostic scanner (Snap-on / Bosch / Ford IDS) concurrently on the vehicle CAN network.
- Confirm zero Diagnostic Trouble Codes (DTCs) generated (specifically U0100 lost communication or CAN-bus error frames).
- Confirm zero acknowledgment pulses (`ACK`) transmitted on `can0` during bus operation.
- [] **2.4 Parasitic Battery Draw & Low-Voltage Disconnect (LVD) Calibration**
- Connect in-line digital multimeter to monitor sleep current.
- Confirm device enters ultra-low power standby (`< 15 mA`) within 90 seconds of vehicle ignition cut-off.
- Validate Low-Voltage Disconnect thresholds:
 - 12V Passenger Systems: Hardware cut-off trips at **11.8V DC**.
 - 24V Commercial Systems: Hardware cut-off trips at **23.6V DC**.

5. Phase 3: WASM Carrier Module Sandbox Deployment (Days 13 to 20)

Inject and test carrier-specific underwriting and claim adjudication logic within the dual-consensus WASM runtime.

- [] **3.1 Carrier WASM Logic Compilation**
- Compile the carrier underwriting rules to `wasm32-unknown-unknown` target.
- Verify binary size is within enterprise envelope (< 2.0 MB).

- Inspect WASM exports: `adjudicate_claim` , `calculate_premium` , `evaluate_parametric_trigger` .
- [] **3.2 Dual-Consensus Engine Execution Audit**
- Execute test batch of 1,000 synthetic collision packets through both Wasmtime (Cranelift) and V8 runtimes.
- Verify bit-for-bit consensus across all output structures: `Engine A (Cranelift): SHA256(Result) = 7f83b165...` `Engine B (V8): SHA256(Result) = 7f83b165...` `Consensus State: MATCH (Status 200 OK)`
- Confirm that an injected floating-point NaN mismatch causes an immediate deterministic sandbox trap.
- [] **3.3 Memory Isolate & Zero-Syscall Enforcement**
- Confirm the WASM linear memory is hard-capped at 64KB (1 page).
- Inject malicious WASM bytecode attempting `socket()` , `open()` , and `fork()` .
- Verify sandbox immediately halts execution with `WasmTrapError: illegal host capability invocation` .

6. Phase 4: Optical Forensics & DPPA Privacy Pipeline (Days 21 to 25)

Validate fraud defense mechanisms and privacy redaction on live imagery and telemetry streams.

- [] **4.1 Optical Forensics & Synthetic Media Rejection Test**
- Feed 50 physical collision photographs and 50 generative AI images (Midjourney v6, Stable Diffusion XL) into `backend/pixel_forensics_engine.py` .
- Verify Bayer CFA demosaicing detection:
 - Physical CMOS photos: Autocorrelation peak at (π, π) yields CFA score > 0.60 (AUTHENTIC).
 - Generative AI photos: Score < 0.40 triggers `SYNTHETIC_MEDIA_FLAGGED` alert.
- Verify execution speed: Complete 5-vector optical scan completes in `< 120 ms` .
- Verify output meets Federal Rule of Evidence 702 (Daubert Standard).
- [] **4.2 Edge PII Sanitization Audit (`backend/pii_sanitizer.py`)**
- Inject raw mock telemetry packets containing full 17-digit VINs, driver Social Security Numbers, and unblurred facial images.
- Inspect output payload exported across the WireGuard tunnel:
 - [x] VIN stripped to 8-character WMI/VDS prefix (e.g., `1HGCR2F8`).
 - [x] SSN replaced with salted HMAC-SHA256 token (`HMAC-TOKEN-XXXX`).
 - [x] GPS coordinates fuzzed to 2 decimal places (`~1.1 km` spatial grid).
 - [x] Camera images blurred over facial bounding boxes (Gaussian $\sigma=15$).
- Confirm adherence to Driver's Privacy Protection Act (**18 U.S.C. § 2721**).
- [] **4.3 Ethan Conversational AI Telephony Validation**
- Test live incoming SIP claim call over UDP port 5060 / RTP 10000-20000.
- Confirm Opus 24 kbps VBR audio codec engagement.
- Validate forensic acoustic vector scan level 21 (60Hz Electric Network Frequency hum matching).
- Verify end-to-end voice turnaround latency is `< 150 ms` .

7. Phase 5: Solvency Automation & Pilot Executive Sign-Off (Days 26 to 30)

Simulate high-severity catastrophic losses and execute formal treaty waterfall cessions.

- [] **5.1 Catastrophe Peril Modeling Validation (Tab 40)**

- Load carrier pilot portfolio exposure into the stochastic simulation engine.
- Run 100,000 Monte Carlo iterations across hurricane and earthquake hazard sets.
- Confirm calculation of Solvency II 99.5% 1-in-200-year Value at Risk (VaR) matches carrier actuarial benchmarks within $\pm 0.05\%$.

- [] **5.2 Reinsurance Waterfall Execution (Tab 41)**

- Submit simulated catastrophic collision cluster totaling \$15,000,000 in gross incurred losses.
- Verify automatic waterfall allocation:
 1. Net Retention: First \$1,000,000 absorbed.
 2. Quota Share (80/20): \$4,000,000 ceded to lead syndicate.
 3. Working XOL (\$10M xs \$5M): \$10,000,000 fully exhausted.

- Confirm automated generation of NAIC Schedule F and Lloyd's MS9 cession bordereau.

- [] **5.3 Parametric Smart Contract Rehearsal (Tab 42)**

- Trigger synthetic USGS ShakeMap (0.45g PGA) and NOAA Hurricane (130 kt sustained wind) dual-oracle alerts.
- Verify state machine transition latency is `< 2.0 ms`.
- Confirm generation of ISO 20022 `pacs.008.001.08` Fedwire dispatch message for \$2,500,000.00 automated settlement.

- [] **5.4 Final Pilot Governance Review & Executive Hand-Off**

- Complete joint review between Carrier CISO, Chief Actuary, Head of SIU, and Ether Terminal LLC deployment team.
- Archive all pilot audit logs into tamper-evident Ed25519-signed verification dossier.
- Sign off on production rollout authorization.

8. Pilot Verification Sign-Off Matrix

Role	Name / Title	Organization	Approval Date	Signature / Enclave Key ID
Carrier Project Sponsor	____	Carrier Corporate	//2026	KEY-SPONSOR- _____
Chief Information Security Officer	____	Carrier IT & Infosec	//2026	KEY-CISO- _____
Chief Actuary / Risk Officer	____	Carrier Actuarial	//2026	KEY-ACTUARIAL- _____
Director of Special Investigations (SIU)	____	Carrier Claims/Fraud	//2026	KEY-SIU- _____
Lead Deployment Architect	____	Ether Terminal LLC	//2026	KEY-ETHER-DEPLOY- _____

